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RWANDA

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SECOND REVIEWS UNDER THE POLICY COORDINATION INSTRUMENT AND THE ARRANGEMENT UNDER THE RESILIENCE AND SUSTAINABILITY FACILITY, REQUEST FOR THE MODIFICATION OF END-DECEMBER 2023 QUANTITATIVE TARGETS, REQUEST FOR REPHASING OF ACCESS UNDER THE RESILIENCE AND SUSTAINABILITY FACILITY, REQUEST FOR A STAND-BY CREDIT FACILITY ARRANGEMENT, AND THE 2023 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS[1]

Approved By

Catherine Pattillo and Martin Čihák (IMF), Manuela Francisco and Hassan Zaman (IDA)

The Debt Sustainability Analysis was prepared jointly by the staff of the International Monetary Fund and the International Development Association.

Rwanda: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Moderate*
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Some space to absorb shocks*
Application of judgment	*No*

The updated Bank/Fund debt sustainability analysis (DSA) for Rwanda continues to indicate a moderate risk of external and overall public debt distress. The current debt-carrying capacity is consistent with a classification of 'strong'.[2] The PV of external debt-to-GDP ratio and of external debt-to-export ratio remain below their indicative thresholds under the baseline scenario and several shock scenarios, though the PV of debt-to-GDP ratio breaches the threshold under the most extreme shock scenario and the historical scenario. The PV of overall PPG debt temporarily breaches the threshold under the most severe scenario. The baseline scenario is based on the macroeconomic projections presented in the accompanying staff report for the SCF request and PCI/RSF second review. Rwanda's financing strategy assumes continued support from bilateral and multilateral development partners over the medium term, with highly concessional loans for new external borrowing under IDA20 and an increasing share of domestic financing in the

[1] This debt sustainability analysis was conducted using the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (LIC-DSF) that was approved in 2017. The fiscal year for Rwanda is from July to June; however, this DSA is prepared on a calendar year basis.

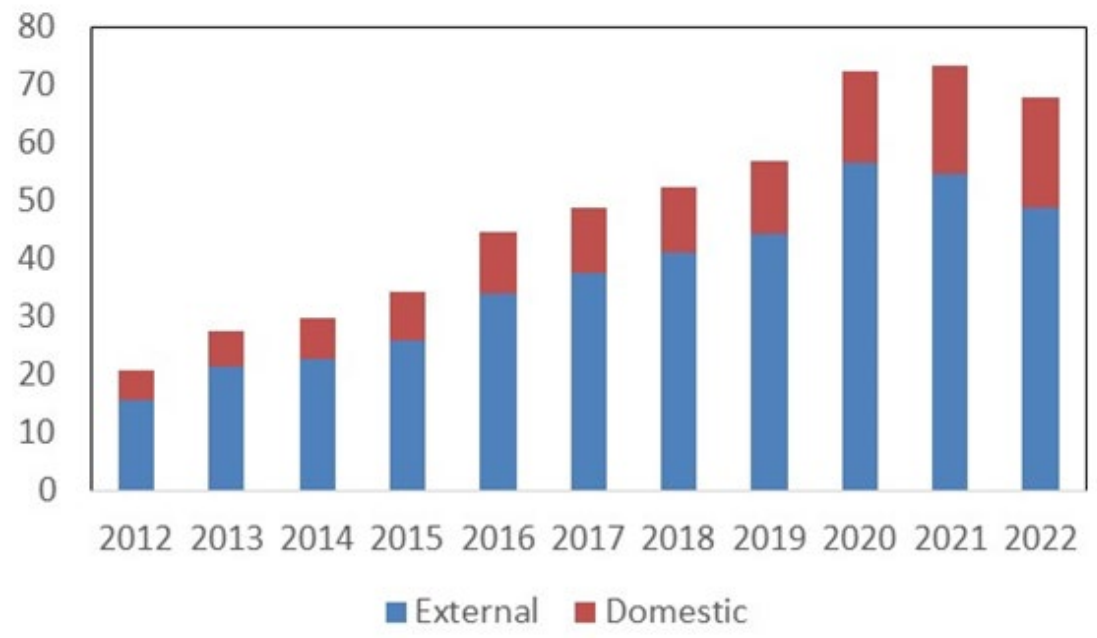
[2] Rwanda has a debt carrying capacity indicator score of 3.18. This implies a classification of strong debt carrying capacity, which is the same classification as under the previous DSA.

long term. Compared to the previous assessment, this DSA incorporates a 14-month SCF with an access level of 125 percent of quota (SDR 200.25 million),partners over the medium term, with highly concessional loans for new external borrowing under IDA20 and an increasing share of domestic financing in the long term. Compared to the previous assessment, this DSA incorporates a 14-month SCF with an access level of 125 percent of quota (SDR 200.25 million), emergency financing from the World Bank totaling US$ 85 million, and rephased RSF disbursements. Key risks include a decline in concessional financing, U.S. monetary policy tightening and U.S. dollar appreciation, and terms-of-trade shocks. Rwanda remains susceptible to adverse market conditions and climate shocks.

BACKGROUND

1. Rwanda's public and publicly-guaranteed (PPG) external debt-to-GDP ratio increased by 33.1 percentage points over the last decade, driven by loans to meet the development needs envisaged in the National Strategy for Transformation (NST), but also to respond to the fallout from the COVID-19 pandemic. The development needs are supported by a comprehensive public investment strategy, including three large projects to support trade and tourism through a series of public-private partnerships and external guarantees outside the budgetary central government (construction of the Kigali Convention Center completed in 2016, the expansion of the national airline RwandAir, and the ongoing construction of the Bugesera airport). These developments contributed to an increase in PPG external debt by 21.8 percentage points in the five years preceding the COVID-19 shock. At the same time, the increase in the fiscal deficit due to revenue shortfalls and a scaling up in spending to address the COVID-19 crisis led to a sharp debt increase by an additional 12.1 percentage points in 2020, followed by 1.9 percentage points decline in 2021. As a result, external PPG debt has risen from 15.6 percent of GDP in 2012 to 48.9 percent in 2022 (Text Figure 1). Total PPG debt stood at 68 percent of GDP at end-2022, 5.4 percentage points (pp) lower than in 2021 on the back of favorable price and exchange rate dynamics. Interest rates on domestic T-bills and T-bonds currently range from 8.3 percent (28 days) to 13.2 percent (20 years).

Text Figure 1. Rwanda: Total PPG Debt
(Percent of GDP)



Sources: Rwandan authorities and IMF Staff Calculations.

Text Table 1. Rwanda: Decomposition of Public and Publicly-Guaranteed Debt and Debt Service by Creditor, 2022–24[1]

	Debt Stock (end of period)			Debt Service					
	2022			2022	2023	2024	2022	2023	2024
	(In US$ mn)	*(Percent total debt)*	*(Percent GDP)* [5]	*(In US$ mn)*			*(Percent GDP)*		
Total	8,648.1	100.0	67.5	796.5	1,232.5	528.6	6.0	8.9	3.8
External	6,193.6	71.6	48.3	181.3	271.4	226.0	1.4	2.0	1.6
Multilateral creditors[2]	4,633.6	53.6	36.2	85.5	129.5	143.5	0.6	0.9	1.0
IMF	451.1	5.2	3.5	1.0	36.8	36.7	0.0	0.3	0.3
World Bank	2,698.1	31.2	21.1	49.8	58.1	70.4	0.4	0.4	0.5
ADB/AfDB/IADB	1,061.0	12.3	8.3	18.9	18.6	20.6	0.1	0.1	0.1
Other Multilaterals	423.4	4.9	3.3	15.8	16.0	15.8	0.1	0.1	0.1
o/w: IFAD	183.2	2.1	1.4	5.2	5.4	5.8	0.0	0.0	0.0
BADEA	80.7	0.9	0.6	0.1	0.1	0.1	0.0	0.0	0.0
Bilateral Creditors	836.4	9.7	6.5	36.2	31.5	37.1	0.3	0.2	0.3
Paris Club	287.8	3.3	2.2	0.3	0.1	0.1	0.0	0.0	0.0
o/w: JICA	142.3	1.6	1.1	0.0	0.0	0.0	0.0	0.0	0.0
AFD	117.4	1.4	0.9	0.2	0.1	0.1	0.0	0.0	0.0
Non-Paris Club	548.6	6.3	4.3	36.0	31.4	37.1	0.3	0.2	0.3
o/w: EXIM-CHINA	334.0	3.9	2.6	16.9	16.2	21.3	0.1	0.1	0.2
SFD	76.0	0.9	0.6	4.3	4.1	4.3	0.0	0.0	0.0
Bonds	680.6	7.9	5.3	38.1	95.8	33.6	0.3	0.7	0.2
Commercial creditors	43.1	0.5	0.3	21.4	14.6	11.8	0.2	0.1	0.1
o/w: Trade Development Bank	39.6	0.5	0.3	14.1	12.3	11.7	0.1	0.1	0.1
o/w: EDC	2.2	0.0	0.0	9.1	2.2	0.0	0.1	0.0	0.0
Other international creditors	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Domestic	2,454.5	28.4	19.2	615.2	961.2	302.5	4.6	6.9	2.2
Held by residents, total	2,454.5	28.4	19.2	615.2	961.2	302.5	4.6	6.9	2.2
Held by non-residents, total	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
T-Bills	425.8	4.9	3.3	438.9	414.7	0.0	3.3	3.0	0.0
Bonds	1,219.8	14.1	9.5	99.4	214.9	206.4	0.7	1.6	1.5
Loans	808.8	9.4	6.3	76.9	331.6	96.1	0.6	2.4	0.7
Memo items:									
Collateralized debt[3]	0.0	0.0	0.0	…	…	…	…	…	…
Contingent liabilities[4]	0.0	0.0	0.0	…	…	…	…	…	…
Nominal GDP (US$ million)	…	…	…	13,309	13,846	13,657	…	…	…

[1]As reported by country authorities according to their classification of creditors, including by official and commercial. Debt coverage is the same as the DSA.

[2]"Multilateral creditors" are simply institutions with more than one official shareholder and may not necessarily align with creditor classification under other IMF policies (e.g. Lending Into Arrears).

[3]Debt is collateralized when the creditor has rights over an asset or revenue stream that would allow it, if the borrower defaults on its payment obligations, to rely on the asset or revenue stream to secure repayment of the debt. Collateralization entails a borrower granting liens over specific existing assets or future receivables to a lender as security against repayment of the loan. Collateral is "unrelated" when it has no relationship to a project financed by the loan. An example would be borrowing to finance the budget deficit, collateralized by oil revenue receipts. See the joint IMF-World Bank note for the G20 "Collateralized Transactions: Key Considerations for Public Lenders and Borrowers" for a discussion of issues raised by collateral.

[4]Guaranteed debt is included in public debt.

[5]Data is reported by the authorities based on calculations in national currency, deviates from U.S. dollar-based calculations due to the difference between end-of-period exchange rate (applied for nominator) and period-average exchange rate (applied for denominator).

Sources: Rwandan authorities and IMF Staff Calculations.

Public Debt Coverage

2. The DSA covers the central government, guarantees, and state-owned enterprises (Text Table 2). The Ministry of Finance and Economic Planning (MINECOFIN) publishes annual debt data, covering domestic and external debt of the central government, broken down by multilateral, bilateral and commercial debt, as well as information on both domestic and external guarantees and domestic and

external debt held by all state-owned enterprises (SOEs).[3] There is no debt stemming from extra budgetary funds, long term central bank financing of the government, nor the state-owned social security fund. The local government debt is also covered but the existing stock to date is marginal,[4] and its contracting is subject to approval by MINECOFIN. The contingent liabilities shock (6.5 percent of GDP) accounts for potential fiscal costs associated with a theoretical banking crisis, as well as fiscal risks of existing public-private partnerships (PPPs). All SOE guaranteed and non-guaranteed debt is included in the baseline.

Text Table 2. Rwanda: Coverage of Public and Publicly-Guaranteed Debt and Parameters for Contingent Liability Shocks for the Tailored Stress Test

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

1	The country's coverage of public debt	The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	0	SOE debt is covered in PPG debt
4	PPP 2/	35 percent of PPP stock	1.5	default, i.e. 35 percent of PPP stock
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
	Total (2+3+4+5) (in percent of GDP)		6.5	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.
2/ When PPP stock is less than 3 percent of GDP, as reflected in the World Bank's database, then test is set to zero.
3/ Based on project value

UNDERLYING ASSUMPTIONS

3. The macroeconomic assumptions underlying the baseline scenario reflect recent economic developments and policies (Text Table 3). The recent devastating floods amplified the underlying imbalances, while the reconstruction costs are projected to be substantial. Based on the authorities' assessment conducted with the support of the World Bank, the reconstruction cost could go up to US$451 million (3 percent of GDP) over the next five years to address the emergency outlays and rebuild climate resilient infrastructures. Compared to the previous DSA, the outturn for 2022 growth was much better than anticipated (8.2 percent vs. 6.8 percent).

- *Growth:* Growth is expected to soften to 6.2 percent and 6.6 percent in 2023 and 2024,[5] respectively, remaining broadly unchanged relative to the first PCI/RSF review in May 2023 as the negative impact of the floods is expected to be offset by the activity boost from reconstruction. Over the medium term, growth projections remain unchanged and economic

[3] Operational leases on RwandAir (SOE) aircrafts and existing public-private partnerships (power purchase agreements, water purchase agreements, etc.) are not part of the PPG debt.

[4] Local government debt stood at RWF 2.4 bn or 0.02 percent GDP at end-2022.

[5] As 2023 and 2024 growth projections reflect potential second round effects of the recent floods, the estimate is different from GDP growth rate in the World Bank MPO (Oct 2023).

activity is expected to be buoyed by private sector-led economic diversification, the pickup in construction of the new airport and the subsequent boost to the services sector. Private consumption and investment are the main growth drivers in the medium term as fiscal consolidation proceeds. Although headline inflation is expected to decline to 6.0 percent in 2024, this forecast is higher than the first review and previous DSA forecast of 5.0 percent, reflecting easing food inflation but elevated core inflation due to continued exchange rate depreciation passthrough. Over the long term (Box 1), growth from the supply-side is expected to be supported by a dynamic service sector, gradual industrialization, and continued significant contributions from agriculture for some period. On the demand side, robust and steady private consumption and investment will support the rebalancing of the economy from public sector-led towards private sector-led growth.

- *Fiscal:* The fiscal consolidation path envisaged in the previous DSA is now more backloaded to accommodate for reconstruction spending in the near term. Given the urgent need for reconstruction from the floods and the limited scope for reprioritization of expenditures in the short-term, a moderate fiscal loosening is warranted to allow for reconstruction efforts. Despite this relaxation, the fiscal impulse in FY23/24 remains sharply negative compared to FY22/23. In the medium term, fiscal policy will be geared towards maintaining fiscal and external sustainability: further domestic revenue mobilization and spending rationalization efforts are expected. These include revenue measures focused on the application of VAT to IT equipment and mobile phones, an increase in the fuel levy, increases in excises and revenue administration through FY25/26, with more comprehensive revenue strategies expected thereafter. In parallel, the authorities' spending rationalization strategy underpinning the medium-term fiscal consolidation path is expected as part of the FY24/25 Budget Framework Paper. Further implementation of the Organic Budget Law, including institutionalizing fiscal risk management practices, as well as PIMA recommendations, including the annual publication of a pipeline of appraised major projects and project selection criteria, will further uphold expenditure efficiency.[6] Overall, the projections continue to assume strong fiscal policy consolidation to accelerate public debt convergence to the anchor.
- *External:* The current account improved in 2022, reflecting a temporary reduction in capital imports, tourism recovery, and strong remittances, but the deficit is projected to widen in 2023 due to the adverse impact of poor agricultural seasons on food imports and flood-related reconstruction efforts. The revised policy-mix underpinning the SCF assumes greater real effective exchange rate depreciation.[7] Over the medium term, the trade balance and current account deficit is projected to gradually improve and close the gap with the

[6] As part of the FY23/24 Policy Performance Actions (PPA) under IDA's sustainable Development Finance Policy (SDFP), the World Bank is supporting authorities with the publication of the pipeline of appraised projects, including project costs and selection criteria of new major projects. Under the same framework, the World Bank supported the approval of the organic law on public finance management, institutionalizing fiscal risk management practices in FY21/22 PPAs.

[7] Exchange rate depreciation is expected to have a greater net effect on import prices, contributing partially to a decline in USD GDP deflator over 2023-24. Higher overall price levels have pushed up deflator levels in both years.

current account norm (see External Sector Assessment), underpinned by further real effective exchange rate adjustment and higher domestic savings from fiscal consolidation efforts.

Text Table 3. Rwanda: Key Macroeconomic and Debt Assumptions—Comparison with the Previous Debt Sustainability Analysis

Calendar year	2022	2023	2028	2033	2038	2042	2029-42
	Actual	Projections					
Selected indicators from the macro-framework and debt data							
(Percent, unless otherwise indicated)							
	PV of PPG External Debt to GDP Ratio						
2022 DSA (PCI-RSF Request)	33.6	35.8	39.2	34.0	27.2	23.4	30.6
2023 DSA (current)	31.2	37.9	37.5	31.4	25.5	21.1	28.4
	PV of Public Debt to GDP Ratio						
2022 DSA (PCI-RSF Request)	52.7	55.0	49.2	47.0	49.8	52.8	48.7
2023 DSA (current)	50.5	54.6	45.3	44.4	46.0	47.8	45.4
	Grant Element of New External Borrowing						
2022 DSA (PCI-RSF Request)	47.9	30.6	39.9	37.0	34.7	32.8	34.4
2023 DSA (current)	39.6	32.8	41.6	36.9	35.3	33.6	35.9
	Real GDP Growth (annual percent change)						
2022 DSA (PCI-RSF Request)	6.8	6.2	7.3	7.0	6.6	6.5	6.8
2023 DSA (current)	8.2	6.2	7.3	7.0	6.6	6.5	6.8
	Current Account Balance (percent of GDP)						
2022 DSA (PCI-RSF Request)	-10.5	-12.4	-9.9	-10.0	-9.2	-8.6	-9.6
2023 DSA (current)	-9.8	-11.5	-7.7	-8.7	-7.9	-6.9	-8.0
	Exports of goods and services (percent of GDP)						
2022 DSA (PCI-RSF Request)	24.1	25.0	25.6	24.6	24.9	26.0	25.0
2023 DSA (current)	26.2	28.8	32.1	29.3	28.4	29.0	29.3
	Fiscal balance (percent of GDP)						
2022 DSA (PCI-RSF Request)	-8.0	-7.4	-2.9	-4.4	-3.8	-3.8	-3.7
2023 DSA (current)	-7.3	-6.2	-3.3	-4.0	-3.8	-3.8	-3.8
	CPI, period average (percent)						
2022 DSA (PCI-RSF Request)	12.6	7.9	5.0	5.0	5.0	5.0	5.0
2023 DSA (current)	13.9	14.5	5.0	5.0	5.0	5.0	5.0

1/ Previous full DSA was completed in November 2022.
Sources: Rwandan authorities; IMF and World Bank staff estimates and projections.

4. Rwanda's financing strategy assumes continued support from bilateral and multilateral development partners over the medium term (Table 5). Besides financing from the World Bank and the IMF, the projection assumes disbursements of external financing from the African Development Bank and several other multilateral and official bilateral partners. In the context of fiscal consolidation and negotiated new external resources, the share of domestic financing is projected to remain low until 2030 and pick up

thereafter.[8] The share of market-based external financing is also projected to increase starting 2030, although very slowly. Under the IDA20 financing terms, the volume of loans is projected to increase given the shift from 50-50 grant-loan financing under IDA19 to 100 percent loans, hence, the fiscal deficit and the nominal debt will increase as well, but given the higher concessional terms of IDA20 loans, the expected impact on the present value of debt is marginal.[9] IDA financing signed after 2024 is assumed to be in the form of 100 percent credit on regular IDA terms.

5. The DSA also takes into account the proposal for a 14-month SCF-supported program with the access level of 125 percent of quota (SDR 200.25 million).[10] The SCF-supported program will help the authorities maintain macroeconomic and external stability in the face of the recent agricultural and flood-related shock. Financial assistance under the SCF, coupled by a recalibration of the policy mix, is projected to support the rebuilding of external buffers. This additional provision of concessional external financing in place of more expensive domestic financing will also improve underlying public debt dynamics.

6. Realism tools indicate that the planned fiscal adjustment is ambitious (Figure 4). A 3-year fiscal consolidation in the primary balance is expected to peak at 4.3 percentage points of GDP from 2024 to 2027. Such adjustment lies in the top quartile of the distribution of past adjustments for a sample of low-income countries (LICs), signaling that the envisaged fiscal adjustment in the baseline scenario is ambitious based on past experiences in LICs. The adjustment reflects the domestic revenue mobilization and spending rationalization measures expected to be implemented during the period covered under the PCI approved in 2022 and the proposed SCF.

7. In the past, PPG debt dynamics have been strongly affected by the materialization of fiscal risks (Figure 3). Changes in total public debt over the past five years have been driven by higher-than-anticipated primary deficits due to the pandemic response and unanticipated developments of the debt outside the budgetary central government, leading to a higher-than-expected debt accumulation of about 20 percentage points of GDP—well in excess of the 75 percent quartile of other LICs. The ongoing efforts to mitigate unanticipated fiscal developments outside the central government, supported by IMF TA, remain a key pillar of the PCI. Going forward, the evolution of public debt will continue to be dominated by the path of the primary fiscal deficit, real GDP growth, real effective exchange rate adjustment, and the availability of concessional financing.

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[8] The domestic market has increased in its capacity to provide financing, from about 60bn Rwf to 300bn Rwf annually in recent years, and with longer bond maturities. Main creditors are domestic banks and pension funds. Though the government still turns to domestic financing as a residual following lower-cost external financing, strategies to reduce the cost of domestic borrowing include re-openings of previously issued longer-term bonds with lower rates than prevailing market rates, while longer-term borrowing helps lower domestic portfolio re-financing risks. The recent successful launch of the Sustainability Linked Bond by the Development Bank of Rwanda, with support from the World Bank, will also contribute to capital market development.

[9] 50-year loans under IDA20 have grant element of about 74 percent, while 50-50 grant-loan financing that prevailed under IDA19 has grant element of about 77 percent. With 3 percentage points difference, if cumulative disbursements under IDA20 reach 3.2 percent GDP, the effect on PV of external debt would increase by about 0.1 percent GDP.

[10] The RSF loan has a 20-year maturity with 10 ½ year grace period, with borrowers paying an interest rate with a modest margin over the three-month SDR rate.

Box 1. Rwanda: Long-term macroeconomic assumptions 1/

Growth. According to World Bank analysis, long term growth projections of 6.5-7 percent are driven by four broad group of fundamentals: investment (public and private), human capital, total factor productivity (TFP) and demographics. The baseline assumptions for each growth driver are designed to capture a plausible path for the variable if present trends continue, taking into account both Rwanda's own history, as well as reasonable values among other peer countries. Given the government's strong historical record on implementing reforms, the baseline also includes "typical" levels of reforms that might be achieved by a typical country in Rwanda's situation. Thus, the baseline assumes that TFP growth in the long run is 1.8 percent, which is between the averages 2000-19 and 2000-14 averages. The productivity projection is based on assumption of continued increase in FDI and trade openness. The baseline also assumes the government will continue its reform efforts to improve human capital indicators. To calculate this human capital path, the analysis identifies 12 countries that in 1980 had the same years of schooling of 20-24 year olds as Rwanda in 2020 (6.5 years ± 0.5 years) – Rwanda's "schooling peers" – and then trace out the gains in education of the schooling peers over the next 10 years. In terms of investments, the baseline assumption is that investment continues at 28 percent of GDP for the whole projection period. While this represents a slowing of the increase in the 2000s, and a less optimistic assumption than in Rwanda's Vision 2050 document, sustaining investment at or above 30 percent of GDP is very difficult among peer countries or even among growth miracle economies. Finally, the population growth is projected to slow from around 2.3 percent currently to about 1.4 percent over the next 30 years, as Rwanda goes through a "demographic transition" with lower fertility rates. Other things equal, the around1ppt slowdown in population growth will slow GDP growth by around half a percentage point in the medium term but boost GDP per capita slightly.

Exports. There is potential for Rwandan exports to increase substantially in medium to long term. First, regional integration via AfCFTA presents an enormous opportunity to boost exports in coming years. To capture this benefit, Rwanda has been a leader driving negotiations of a series of protocols to the AfCFTA Treaty with clear, ambitious, and enforceable rules and disciplines, adopting legislative changes, and building implementation and enforcement capacity. Second, Rwanda has achieved considerable growth in exports of minerals and metals (minerals exports increased more than 10 times from 2010 to 2020), and further growth is expected with recent investments in the sector. Rwanda's low labor cost could help achieve further increases in exports of textiles and clothing, supported by investments made in recent years. The potential for increasing services exports also is significant. Rwanda has revealed comparative advantage indices of greater than one for government goods and services, construction, travel and transport services. Construction companies in Rwanda have the skills required to operate in other regional countries, for example in building and operating power plants. Rwanda's booming tourism sector have already recovered from the decline during the pandemic and is expected continue its expansion, as the country maintains a strong focus on the MICE strategy.

1/ Based on World Bank analysis from the ongoing Rwanda Country Economic Memorandum

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

8. Rwanda's debt-carrying capacity continues to be assessed as "strong", unchanged from the previous DSA (Text Tables 4a and 4b). The composite index (CI) for Rwanda, which measures the debt-carrying capacity in the new LIC-DSF, stands at 3.18, based on macroeconomic data from the October 2023 WEO and from the 2022 Country Policy and Institutional Assessment (CPIA) of the World Bank. This score is above the cut-off value of 3.05 for strong capacity countries.

Text Table 4a. Rwanda: Debt-Carrying Capacity

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.39	4.06	1.56	49%
Real growth rate (in percent)	2.72	6.84	0.19	6%
Import coverage of reserves (in percent)	4.05	39.25	1.59	50%
Import coverage of reserves^2 (in percent)	-3.99	15.40	-0.61	-19%
Remittances (in percent)	2.02	3.14	0.06	2%
World economic growth (in percent)	13.52	2.89	0.39	12%
CI Score			3.18	100%
CI rating			Strong	

New framework	Cut-off values		
Weak	CI ≤	2.69	
Medium	2.69	< CI ≤	3.05
Strong	CI >	3.05	

Text Table 4b. Rwanda: Applicable Thresholds and Benchmarks

Reference: Thresholds by Classiciation

EXTERNAL debt burden thresholds	Weak	Medium	Strong
PV of debt in % of			
Exports	140	180	240
GDP	30	40	55
Debt service in % of			
Exports	10	15	21
Revenue	14	18	23

TOTAL public debt benchmark	Weak	Medium	Strong
PV of total public debt in percent of GDP	35	55	70

9. Rwanda's gross financing needs remain significant, but liquidity risks should be manageable (Figure 5). Market-financing stress tests were applied to Rwanda. Risk indicators from the stress test show that the country has significant gross financing needs (GFNs) and its spread has increased following the recent rise of sovereign spreads globally. However, both GFNs and spreads are below their relevant thresholds.

DEBT SUSTAINABILITY ANALYSIS

External Debt

10. External debt indicators under the baseline remain below their respective thresholds but some breaches would occur under the most extreme shocks (Tables 1 and 3; Figure 1). Solvency indicators, PV of external debt-to-GDP ratio and of external debt-to-export ratio, remain below their indicative thresholds under the baseline scenario and several shock scenarios. However, the PV of external

debt-to-GDP ratio would breach the threshold under the most extreme shock scenario in 2026, and in 2030 under the historical scenario. Liquidity indicators show that liquidity risks are remote, as the authorities' debt management strategy to smooth out the debt servicing profile helped to mitigate the liquidity risks. The debt service-to-exports and debt service-to-revenue ratios remain below their thresholds until 2030 under the baseline and all stress tests applied, but prudent debt management strategy would be needed to mitigate the risks associated with the repayment of the existing Eurobond in 2031.

11. The PV of external debt-to-GDP ratio increases steadily under the historical scenario since the latter assumes large external shock and imbalance reflecting averages of several large shocks and imbalances observed in the past (Table 3 and Figure 1). This is primarily due to the large current account deficit and negative USD GDP deflator calibrated using historical averages, which covered a period including several large shocks (commodity prices and drought) and large external imbalances corrected over 2015–17. Thus, policy adjustment to ensure a steady narrowing of the current account deficit, as envisaged under the baseline scenario (Text Table 2), is key to strengthen robustness of the debt dynamics.

Public Debt

12. Under the baseline scenario, the PPG debt is expected to return to the program debt anchor under the PCI of 65 percent of GDP by 2031, supported by a large, but growth-friendly fiscal consolidation and strong economic activity (Table 2). The nominal PPG debt is projected to peak at 78 percent GDP in 2024 then gradually converge to the debt anchor of 65 percent GDP by 2031.[11] The present value (PV) of PPG debt is projected to decline and achieve the East African Community debt convergence criterion of 50 percent by 2026. These dynamics would require a cumulative reduction in the primary fiscal deficit by 4.3 percentage points of GDP between 2024 and 2026.

13. Contingent on the projected growth trajectory and ambitious fiscal consolidation, the PV of PPG debt stays well below the indicative benchmark of 70 percent of GDP but would nearly reach the benchmark under the growth shock scenario in 2033 and temporarily breach it under the extreme natural disaster shock scenario in 2024 (Tables 2 and 4; Figure 2). In the baseline scenario, the PV of PPG debt peaks at 54.6 percent in 2024 and declines to below 50 percent in 2027. The PV of PPG debt in the growth shock scenario would reach 69 percent in 2033, just below the 70 percent threshold. In the most severe scenario—the natural disaster shock—the PV of PPG debt crosses the 70 percent threshold briefly in 2024 before falling back below it the following year as it steadily declines. The authorities climate-related reform measures, including under the RSF, and priorities in climate-related investment should help reduce variability to climate shocks and strengthen resilience. While liquidity risks are muted in the baseline, reflecting a smooth debt service path achieved by the Eurobond issuance, debt service costs would remain elevated under the severe growth and extreme natural disaster shocks.[12]

[11] For discussion of the debt anchor see Country Report No. 2021/1 and the accompanying staff report for new PCI and RSF request (EBM/22).

[12] The natural disaster stress test was informed by the once-in-100-years flooding scenario discussed in the World Bank's (2022) Country Climate and Development Report and significantly more severe by recent flooding events in the second quarter of 2023. Damages are assumed to reach 11.2 percent of the physical capital, requiring

(continued)

ASSESSMENT

14. Rwanda's debt is assessed to be sustainable with a moderate risk of external and public debt distress. Mechanical ratings suggest a moderate risk of external debt distress with some space to absorb shocks, as well as a moderate risk of overall debt distress, as debt indicators breach some of their respective thresholds under the most extreme standard shock scenarios. Given this assessment of moderate risk of debt distress, the limit on the stock of new external PPG debt will continue to be monitored under the PCI and the new SCF.

15. Against a background of a volatile and challenging external environment, risks to the debt outlook and sustainability could elevate should the situation deteriorate further. The macroeconomic framework which underpins this DSA reflects currently available information. The baseline scenario assumes Rwanda continues to achieve robust growth over the medium term, while concessional financing is expected to decline only gradually in the long term. As the situation evolves, including with respect to the war in Ukraine, commodity prices, foreign inflation and growth, the debt risk assessment might change.

Authorities' Views

16. The authorities broadly agree with the results of the DSA and the overall assessment of a moderate risk of external and overall debt distress. The authorities' debt management strategy continues to be based on maximizing external concessional financing to minimize debt servicing pressures, remaining committed to revenue and spending measures agreed under the PCI and underpinning the medium-term fiscal consolidation path, and bringing debt to the anchor of 65 percent of GDP. Their extension of maturities in domestic borrowing and re-openings of longer-term bonds have helped lower domestic portfolio re-financing risks and reduce the cost of new domestic borrowing. They recognize potential fiscal risks stemming from PPAs and WPAs and continue to strengthen their capacity to manage such risks, including through coverage in the fiscal risk statement. The authorities also highlighted that the IMF-World Bank Debt Sustainability Framework for Low Income Countries would benefit from better incorporating new and evolving climate financing instruments and improved adaptation to climate chocks resulting from climate investments.

17.9 percent of GDP in external financing to be replaced (with capital-to-GDP estimation at 1.6 based on the perpetual inventory method). GDP and exports would each be expected to decline by 4.4 percent.

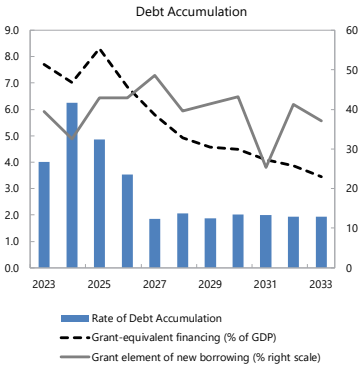
Table 1. Rwanda: External Debt Sustainability Framework, Baseline Scenario, 2022–43

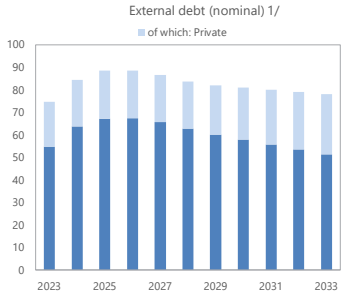
(In percent of GDP, Unless Otherwise Indicated)

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2022	2023	2024	2025	2026	2027	2028	2033	2043	Historical	Projections
External debt (nominal) 1/	**68.1**	**74.5**	**84.2**	**88.4**	**88.5**	**86.6**	**83.6**	**78.1**	**69.3**	**53.2**	**82.3**
of which: public and publicly guaranteed (PPG)	**48.9**	**54.7**	**63.6**	**67.1**	**67.4**	**65.6**	**62.7**	**51.4**	**32.2**	**38.7**	**59.9**
Change in external debt	-8.5	6.5	9.7	4.1	0.2	-1.9	-3.0	-0.9	-1.4		
Identified net debt-creating flows	**-6.2**	**4.4**	**3.0**	**0.0**	**-0.5**	**-1.4**	**-1.8**	**-0.7**	**-2.4**	**5.4**	**-0.2**
Non-interest current account deficit	**8.8**	**10.2**	**10.1**	**8.2**	**7.9**	**7.0**	**5.7**	**6.8**	**4.4**	**10.0**	**7.4**
Deficit in balance of goods and services	14.9	14.9	14.6	14.6	13.3	11.3	10.0	9.4	6.2	**15.4**	**11.4**
Exports	22.5	26.2	28.8	31.5	31.6	32.7	33.2	29.7	29.0		
Imports	37.4	41.2	43.4	46.0	44.8	44.0	43.2	39.1	35.2		
Net current transfers (negative = inflow)	-7.4	-6.4	-6.2	-7.9	-6.9	-6.0	-6.0	-5.0	-3.6	**-6.8**	**-6.1**
of which: official	-6.3	-5.7	-4.9	-5.8	-5.1	-4.2	-4.1	-2.8	-1.6		
Other current account flows (negative = net inflow)	1.3	1.6	1.7	1.5	1.5	1.6	1.8	2.4	1.8	**1.4**	**2.1**
Net FDI (negative = inflow)	**-3.0**	**-3.0**	**-3.8**	**-4.5**	**-4.6**	**-4.5**	**-3.7**	**-4.4**	**-4.9**	**-2.7**	**-4.2**
Endogenous debt dynamics 2/	**-12.0**	**-2.8**	**-3.3**	**-3.7**	**-3.9**	**-3.9**	**-3.8**	**-3.1**	**-2.0**		
Contribution from nominal interest rate	1.0	1.3	1.8	2.0	2.1	2.1	2.0	1.9	2.3		
Contribution from real GDP growth	-5.2	-4.1	-5.0	-5.7	-6.0	-6.0	-5.8	-5.1	-4.2		
Contribution from price and exchange rate changes	-7.8	...	...	...	...	...	...	...	...		
Residual 3/	**-2.4**	**2.0**	**6.7**	**4.1**	**0.7**	**-0.6**	**-1.2**	**-0.2**	**1.0**	**-0.6**	**1.1**
of which: exceptional financing /9	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**28.4**	**31.2**	**37.9**	**41.5**	**41.9**	**40.6**	**39.2**	**32.7**	**21.2**		
PV of PPG external debt-to-exports ratio	**126.4**	**118.8**	**131.8**	**131.9**	**132.9**	**124.2**	**118.1**	**110.1**	**73.0**		
PPG debt service-to-exports ratio	**7.3**	**8.3**	**5.5**	**6.3**	**7.7**	**8.9**	**8.6**	**7.5**	**7.0**		
PPG debt service-to-revenue ratio	**9.0**	**12.2**	**8.8**	**10.3**	**12.2**	**14.2**	**13.7**	**10.0**	**8.1**		
Gross external financing need (Billion of U.S. dollars)	1.3	1.7	1.5	1.2	1.3	1.4	1.4	2.5	5.7		
Key macroeconomic assumptions											
Real GDP growth (in percent)	8.2	6.2	6.6	7.0	7.3	7.3	7.3	7.0	6.5	**6.3**	**7.0**
GDP deflator in US dollar terms (change in percent)	11.3	-2.1	-7.5	-3.6	0.1	0.6	1.4	2.0	2.0	**-0.4**	**-0.1**
Effective interest rate (percent) 4/	1.5	2.0	2.3	2.5	2.6	2.5	2.5	2.7	3.5	**2.9**	**2.5**
Growth of exports of G&S (US dollar terms, in percent)	41.9	21.3	8.3	12.7	7.7	11.7	10.4	7.3	9.3	**11.2**	**9.7**
Growth of imports of G&S (US dollar terms, in percent)	28.0	14.5	4.0	9.4	4.6	5.9	6.8	6.8	8.2	**8.0**	**7.3**
Grant element of new public sector borrowing (in percent)	...	39.4	32.6	42.9	43.0	48.5	39.7	37.1	33.3	**...**	**39.5**
Government revenues (excluding grants, in percent of GDP)	18.1	17.9	18.2	19.3	19.8	20.5	20.8	22.3	25.0	**12.6**	**20.5**
Aid flows (in Billion of US dollars) 5/	0.8	1.3	1.0	1.4	1.2	1.1	1.0	1.1	1.7		
Grant-equivalent financing (in percent of GDP) 6/	...	7.7	7.0	8.3	6.9	5.8	4.9	3.5	2.2	**...**	**5.6**
Grant-equivalent financing (in percent of external financing) 6/	...	63.8	52.7	63.8	65.4	69.6	64.2	59.6	53.2	**...**	**61.0**
Nominal GDP (Billion of US dollars)	13	14	14	14	15	16	18	28	64		
Nominal dollar GDP growth	20.4	4.0	-1.4	3.2	7.4	7.9	8.7	9.1	8.6	**5.8**	**6.9**
Memorandum items:											
PV of external debt 7/	47.6	51.0	58.6	62.8	63.1	61.6	60.1	59.4	58.2		
In percent of exports	211.7	194.2	203.4	199.5	199.9	188.5	181.1	199.9	200.7		
Total external debt service-to-exports ratio	17.1	17.7	15.8	15.9	17.4	18.5	18.2	22.4	32.1		
PV of PPG external debt (in Billion of US dollars)	3.8	4.3	5.2	5.8	6.3	6.6	7.0	9.0	13.5		
(PVt-PVt-1)/GDPt-1 (in percent)		4.0	6.3	4.9	3.5	1.8	2.1	1.9	1.1		
Non-interest current account deficit that stabilizes debt ratio	17.4	3.8	0.4	4.1	7.8	8.9	8.7	7.7	5.8		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

9/ Grant for debt relief under CCRT

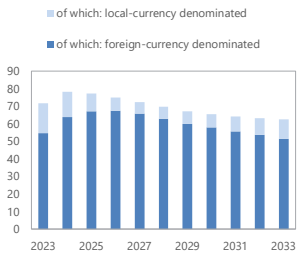
Table 2. Rwanda: Public Sector Debt Sustainability Framework, Baseline Scenario, 2022–43

(In percent of GDP, Unless Otherwise Indicated)

	Actual	Projections								Average 6/	
	2022	2023	2024	2025	2026	2027	2028	2033	2043	Historical	Projections
Public sector debt 1/	**68.0**	**71.6**	**78.0**	**77.2**	**74.8**	**72.3**	**69.5**	**62.4**	**58.7**	**50.8**	**69.6**
of which: external debt	**48.9**	**54.7**	**63.6**	**67.1**	**67.4**	**65.6**	**62.7**	**51.4**	**32.2**	**38.7**	**59.9**
Change in public sector debt	-5.4	3.6	6.4	-0.7	-2.4	-2.6	-2.8	-0.8	0.1		
Identified debt-creating flows	**-4.9**	**3.1**	**5.6**	**-1.2**	**-2.8**	**-2.9**	**-2.8**	**-0.8**	**0.1**	**0.4**	**-0.7**
Primary deficit	**5.4**	**3.9**	**5.3**	**1.7**	**1.0**	**1.0**	**1.9**	**3.0**	**2.3**	**2.4**	**2.6**
Revenue and grants	23.9	22.8	22.1	24.0	24.0	23.9	23.9	24.3	26.2	**18.1**	**23.7**
of which: grants	5.8	4.8	4.0	4.8	4.1	3.4	3.1	2.1	1.2		
Primary (noninterest) expenditure	29.4	26.7	27.4	25.8	25.0	25.0	25.8	27.3	28.5	**20.5**	**26.3**
Automatic debt dynamics	**-10.3**	**-0.8**	**0.5**	**-2.8**	**-3.7**	**-3.8**	**-4.6**	**-3.7**	**-2.2**		
Contribution from interest rate/growth differential	-6.9	-4.7	-3.7	-3.8	-4.8	-4.5	-4.5	-3.8	-2.2		
of which: contribution from average real interest rate	-1.3	-0.7	0.8	1.2	0.5	0.6	0.4	0.4	1.4		
of which: contribution from real GDP growth	-5.5	-4.0	-4.5	-5.1	-5.3	-5.1	-4.9	-4.1	-3.6		
Contribution from real exchange rate depreciation	-3.5	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**-0.2**	**-0.1**	**-0.1**	**-0.1**	**-0.1**	**0.0**	**0.0**	**0.0**	**-0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (use of earmarked fund)	0.0	0.0	-0.2	-0.1	-0.1	-0.1	-0.1	0.0	0.0		
Residual	**-0.5**	**4.3**	**5.0**	**1.5**	**1.5**	**1.0**	**-0.1**	**0.0**	**0.0**	**4.4**	**1.2**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**48.7**	**50.5**	**54.6**	**52.7**	**50.3**	**48.1**	**46.6**	**44.2**	**47.9**		
PV of public debt-to-revenue and grants ratio	**203.5**	**221.8**	**246.8**	**219.2**	**210.0**	**201.1**	**194.9**	**181.7**	**182.8**		
Debt service-to-revenue and grants ratio 3/	**38.6**	**40.1**	**35.7**	**31.3**	**23.7**	**21.1**	**22.4**	**31.9**	**46.4**		
Gross financing need 4/	14.7	13.0	13.0	9.1	6.5	6.0	7.1	10.7	14.5		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	8.2	6.2	6.6	7.0	7.3	7.3	7.3	7.0	6.5	**6.3**	**7.0**
Average nominal interest rate on external debt (in percent)	1.6	1.3	1.5	1.8	1.8	1.8	1.7	1.6	1.8	**2.1**	**1.6**
Average real interest rate on domestic debt (in percent)	-6.7	-0.9	5.9	7.7	7.7	7.8	7.5	6.3	6.1	**2.1**	**6.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	-6.9	...	...	...	...	...	...	...	...	**2.3**	**...**
Inflation rate (GDP deflator, in percent)	16.0	10.4	6.2	5.0	5.0	5.0	5.0	5.0	5.0	**4.8**	**5.6**
Growth of real primary spending (deflated by GDP deflator, in percent)	53.5	-3.5	9.7	0.6	3.9	7.3	10.7	7.6	7.0	**13.8**	**6.4**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	10.8	0.3	-1.1	2.5	3.4	3.6	4.7	3.8	2.2	**-0.1**	**3.1**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government plus social security and extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt. Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

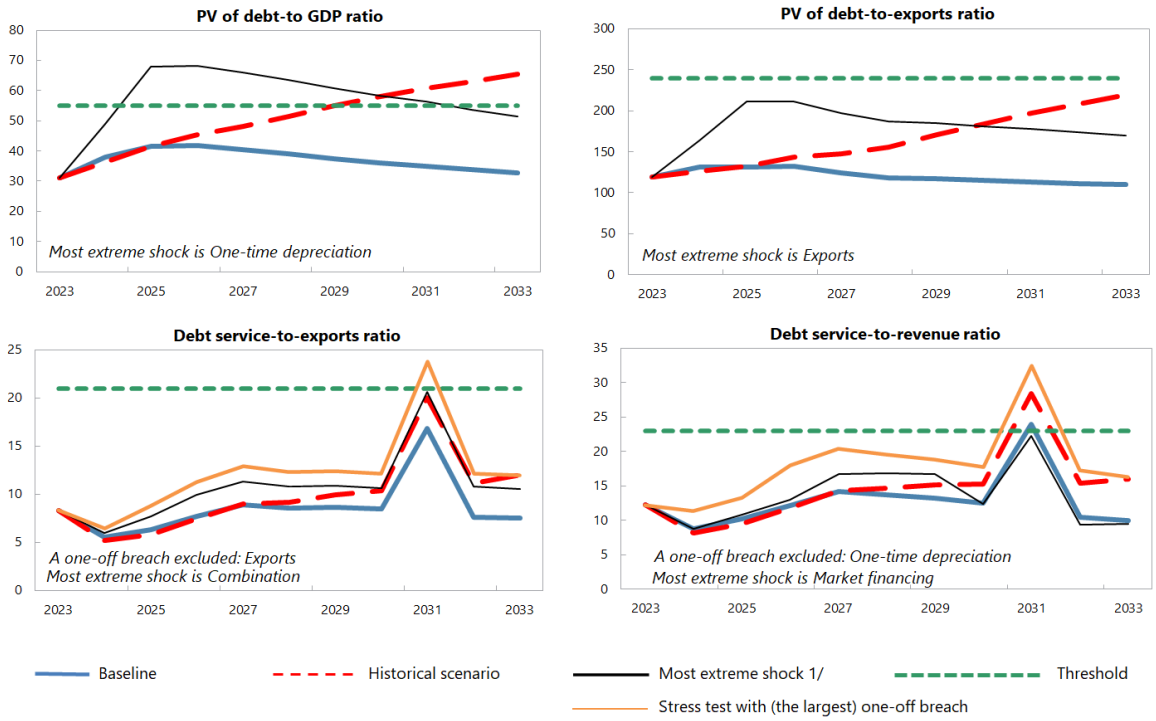
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Rwanda: Indicators of Public and Publicly-Guaranteed External Debt Under Alternative Scenarios, 2023–33 1/ 2/



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	Yes	Yes
Commodity Prices 2/	n.a.	n.a.
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.3%	2.3%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	28	28
Avg. grace period	6	6

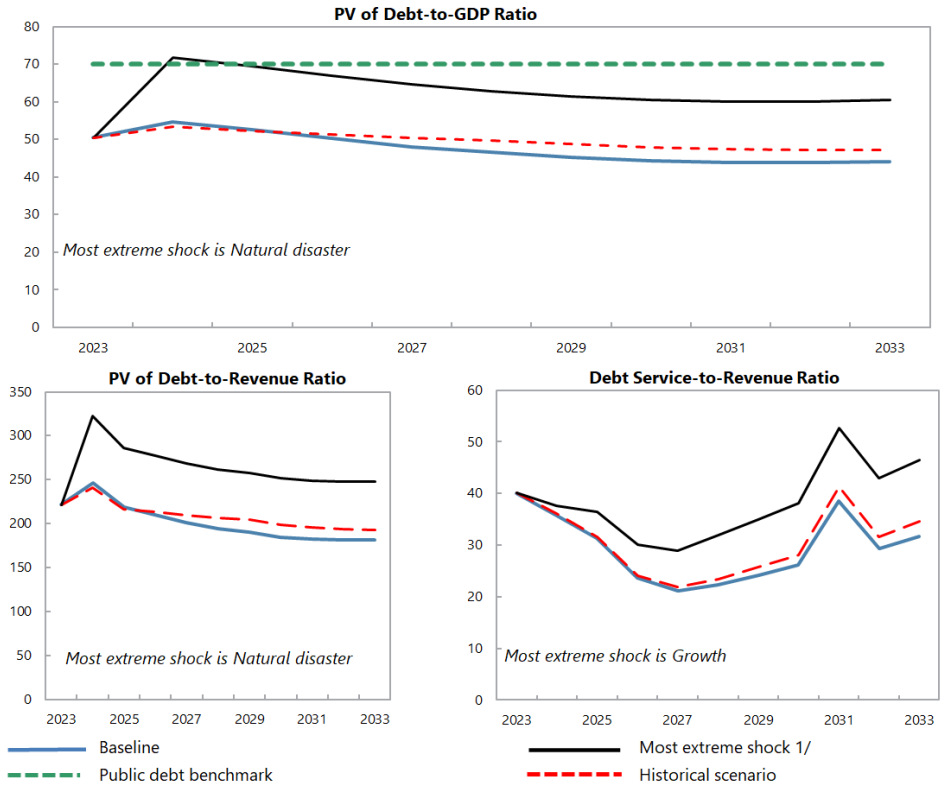
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2033. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Rwanda: Indicators of Public Debt Under Alternative Scenarios, 2023–33 1/



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	63%	63%
Domestic medium and long-term	13%	13%
Domestic short-term	24%	24%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.3%	2.3%
Avg. maturity (incl. grace period)	28	28
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	8.5%	8.5%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	4%	4.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2033. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Rwanda: Sensitivity Analysis for Key Indicators of Public and Publicly-Guaranteed External Debt, 2023–33

(In Percent)

	Projections										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of debt-to GDP ratio											
Baseline	**31.2**	**37.9**	**41.4**	**41.9**	**40.5**	**39.2**	**37.5**	**36.1**	**34.9**	**33.7**	**32.7**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2043 1/	31.2	36.4	41.5	45.4	48.2	51.5	54.9	58.0	60.7	63.0	65.3
B. Bound Tests											
B1. Real GDP growth	31.2	40.7	47.9	48.4	46.9	45.3	43.4	41.8	40.3	39.0	37.8
B2. Primary balance	31.2	38.6	43.9	44.4	43.1	41.8	40.0	38.6	37.2	36.0	34.8
B3. Exports	31.2	41.2	50.6	50.8	49.1	47.4	45.3	43.6	41.9	40.1	38.5
B4. Other flows 2/	31.2	40.0	46.1	46.4	44.9	43.3	41.4	39.9	38.4	36.9	35.5
B6. One-time 30 percent nominal depreciation	31.2	49.1	68.0	68.1	65.9	63.5	60.7	58.4	56.2	53.7	51.4
B6. Combination of B1-B5	31.2	42.8	55.4	55.6	53.8	51.9	49.7	47.8	45.9	43.9	42.1
C. Tailored Tests											
C1. Combined contingent liabilities	31.2	40.8	45.0	45.6	44.5	43.1	41.3	39.9	38.5	37.2	36.0
C2. Natural disaster	31.2	47.8	53.9	55.1	54.8	53.8	52.3	51.0	49.8	48.7	47.7
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	31.2	42.8	47.0	47.7	46.2	44.5	42.3	40.7	39.4	38.1	36.9
Threshold	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**	**55**
PV of debt-to-exports ratio											
Baseline	**118.8**	**131.6**	**131.8**	**132.7**	**124.0**	**117.9**	**116.9**	**114.6**	**113.1**	**111.5**	**109.9**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2043 1/	118.8	126.4	132.1	143.9	147.5	155.1	171.1	183.8	196.8	208.4	219.7
B. Bound Tests											
B1. Real GDP growth	118.8	131.6	131.8	132.7	124.0	117.9	116.9	114.6	113.1	111.5	109.9
B2. Primary balance	118.8	134.0	139.5	140.8	131.9	125.8	124.8	122.4	120.8	118.9	117.0
B3. Exports	118.8	163.6	210.9	211.0	197.1	187.1	185.2	181.3	178.1	173.9	169.8
B4. Other flows 2/	118.8	138.9	146.5	147.0	137.3	130.5	129.2	126.5	124.5	122.0	119.6
B6. One-time 30 percent nominal depreciation	118.8	131.6	167.0	166.8	155.7	147.8	146.3	143.1	140.9	137.2	133.5
B6. Combination of B1-B5	118.8	152.8	161.6	185.7	173.5	164.8	163.1	159.7	156.9	153.1	149.3
C. Tailored Tests											
C1. Combined contingent liabilities	118.8	141.6	143.1	144.4	136.0	129.8	128.8	126.4	124.8	123.0	121.1
C2. Natural disaster	118.8	165.8	171.2	174.4	167.5	162.0	162.8	161.6	161.4	160.9	160.2
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	118.8	131.9	132.5	134.0	125.2	118.7	116.9	114.4	113.1	111.7	110.1
Threshold	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**	**240**
Debt service-to-exports ratio											
Baseline	**8.3**	**5.5**	**6.3**	**7.7**	**8.9**	**8.6**	**8.6**	**8.5**	**16.9**	**7.6**	**7.5**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2043 1/	8.3	5.2	5.8	7.4	9.0	9.2	9.9	10.4	20.0	11.2	12.0
B. Bound Tests											
B1. Real GDP growth	8.3	5.5	6.3	7.7	8.9	8.6	8.6	8.5	16.9	7.6	7.5
B2. Primary balance	8.3	5.5	6.5	8.1	9.3	8.9	9.0	8.8	17.3	8.1	8.0
B3. Exports	8.3	6.4	8.8	11.3	12.9	12.3	12.4	12.1	23.7	12.2	11.9
B4. Other flows 2/	8.3	5.5	6.5	8.1	9.3	8.9	9.0	8.8	17.5	8.4	8.3
B6. One-time 30 percent nominal depreciation	8.3	5.5	6.3	8.7	9.9	9.5	9.5	9.3	17.7	9.7	9.5
B6. Combination of B1-B5	8.3	5.9	7.7	9.9	11.3	10.8	10.9	10.6	20.7	10.8	10.6
C. Tailored Tests											
C1. Combined contingent liabilities	8.3	5.5	6.6	8.0	9.3	8.9	9.0	8.8	17.2	7.9	7.8
C2. Natural disaster	8.3	5.8	7.4	9.0	10.3	10.0	10.2	10.0	18.8	9.1	9.0
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	8.3	5.5	6.6	8.2	10.5	10.5	10.9	8.4	15.7	6.8	7.1
Threshold	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**	**21**
Debt service-to-revenue ratio											
Baseline	**12.2**	**8.8**	**10.3**	**12.2**	**14.2**	**13.7**	**13.2**	**12.4**	**23.9**	**10.4**	**10.0**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2043 1/	12.2	8.2	9.5	11.8	14.4	14.6	15.1	15.2	28.4	15.3	16.0
B. Bound Tests											
B1. Real GDP growth	12.2	9.4	11.9	14.1	16.4	15.8	15.3	14.4	27.6	12.0	11.6
B2. Primary balance	12.2	8.8	10.6	12.8	14.8	14.2	13.7	12.9	24.5	11.1	10.7
B3. Exports	12.2	8.9	10.9	13.7	15.6	15.0	14.4	13.6	25.7	12.8	12.1
B4. Other flows 2/	12.2	8.8	10.6	12.9	14.9	14.3	13.8	13.0	24.8	11.6	11.1
B6. One-time 30 percent nominal depreciation	12.2	11.4	13.3	18.0	20.4	19.5	18.8	17.7	32.4	17.2	16.3
B6. Combination of B1-B5	12.2	9.2	11.9	15.0	17.1	16.4	15.7	14.8	27.8	14.1	13.4
C. Tailored Tests											
C1. Combined contingent liabilities	12.2	8.8	10.7	12.7	14.7	14.2	13.7	12.9	24.4	10.9	10.4
C2. Natural disaster	12.2	8.8	11.6	13.7	15.8	15.3	14.9	14.1	25.5	12.0	11.6
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	12.2	8.8	10.9	13.0	16.7	16.8	16.7	12.4	22.3	9.3	9.5
Threshold	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**	**23**

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Includes official and private transfers and FDI.

Table 4. Rwanda: Sensitivity Analysis for Key Indicators of Public Debt, 2023–33

(In Percent)

	Projections										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of Debt-to-GDP Ratio											
Baseline	**50.5**	**54.6**	**52.6**	**50.2**	**48.1**	**46.5**	**45.3**	**44.2**	**43.9**	**43.8**	**44.1**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2043 1/	50	53	52	51	50	50	49	48	47	47	47
B. Bound Tests											
B1. Real GDP growth	50	59	64	63	63	63	63	64	66	67	69
B2. Primary balance	50	56	56	53	51	49	48	47	46	46	46
B3. Exports	50	58	61	58	56	54	52	51	50	50	49
B4. Other flows 2/	50	57	57	55	53	51	49	48	47	47	47
B6. One-time 30 percent nominal depreciation	50	57	54	50	46	43	41	39	37	36	35
B6. Combination of B1-B5	50	54	54	52	50	49	48	47	47	47	48
C. Tailored Tests											
C1. Combined contingent liabilities	50	60	57	55	52	51	49	48	47	47	47
C2. Natural disaster	50	72	69	67	65	63	62	60	60	60	60
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	50	55	53	51	48	47	45	44	44	44	44
Public debt benchmark	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**	**70**
PV of Debt-to-Revenue Ratio											
Baseline	**221.8**	**246.6**	**219.0**	**209.7**	**200.8**	**194.5**	**190.6**	**184.9**	**182.4**	**181.1**	**181.2**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2043 1/	222	241	217	213	209	206	204	199	196	193	193
B. Bound Tests											
B1. Real GDP growth	222	265	258	257	257	258	263	264	269	274	281
B2. Primary balance	222	252	233	223	213	206	202	195	192	190	190
B3. Exports	222	260	254	244	234	226	221	213	209	205	203
B4. Other flows 2/	222	257	239	229	219	212	208	201	197	194	193
B6. One-time 30 percent nominal depreciation	222	263	227	210	195	183	174	163	155	149	146
B6. Combination of B1-B5	222	245	225	218	211	206	203	198	196	196	196
C. Tailored Tests											
C1. Combined contingent liabilities	222	270	239	229	219	212	207	200	197	195	194
C2. Natural disaster	222	322	286	277	268	261	258	252	249	247	248
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	222	247	220	211	202	196	191	185	182	181	181
Debt Service-to-Revenue Ratio											
Baseline	**40.1**	**35.7**	**31.3**	**23.6**	**21.1**	**22.4**	**24.2**	**26.2**	**38.6**	**29.3**	**31.8**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2043 1/	40	36	32	24	22	23	26	28	41	32	35
B. Bound Tests											
B1. Real GDP growth	40	38	36	30	29	32	35	38	53	43	46
B2. Primary balance	40	36	33	26	23	24	26	27	40	30	33
B3. Exports	40	36	32	25	22	23	25	27	40	31	34
B4. Other flows 2/	40	36	32	24	22	23	25	27	39	30	33
B6. One-time 30 percent nominal depreciation	40	34	31	24	22	23	25	27	40	29	31
B6. Combination of B1-B5	40	35	32	26	24	25	27	30	42	32	35
C. Tailored Tests											
C1. Combined contingent liabilities	40	36	38	26	25	25	26	28	40	30	33
C2. Natural disaster	40	37	53	33	35	31	31	33	45	35	38
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	40	36	32	24	23	25	27	26	37	28	31

Sources: Country authorities; and staff estimates and projections.

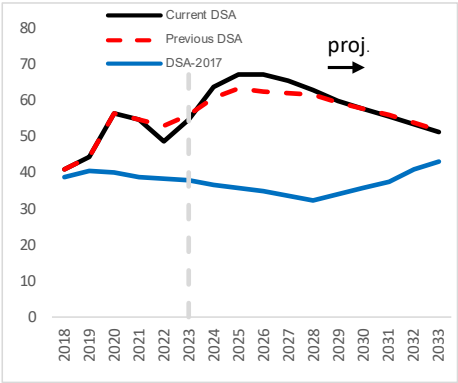
1/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

2/ Includes official and private transfers and FDI.

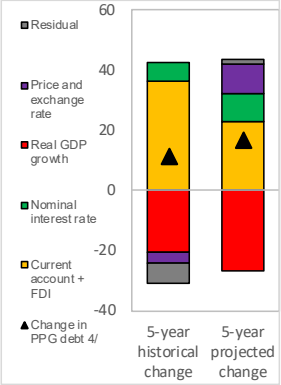
Figure 3. Rwanda: Drivers of Debt Dynamics—Baseline Scenario[1/]

External debt

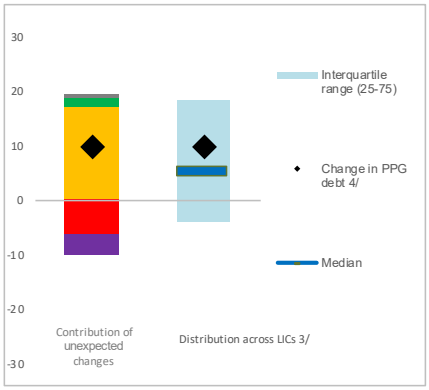
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

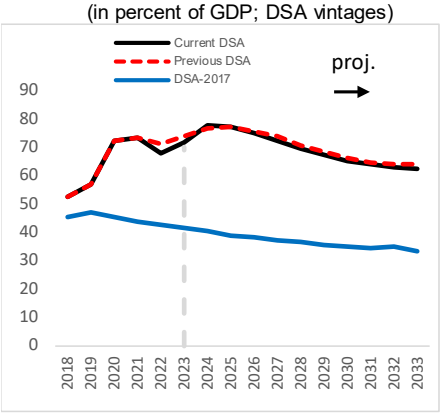


Unexpected Changes in Debt 2/
(past 5 years, percent of GDP)

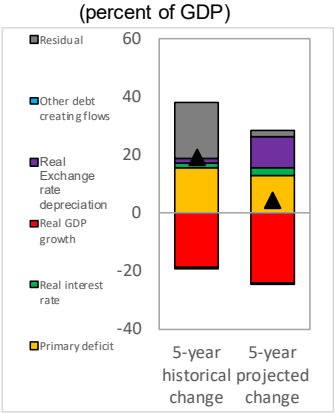


Public debt

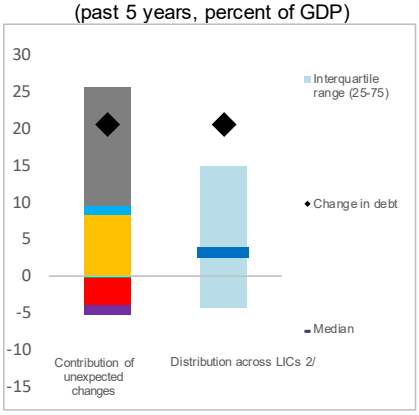
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Compared to 2017 DSA and the previous full DSA in 2022.

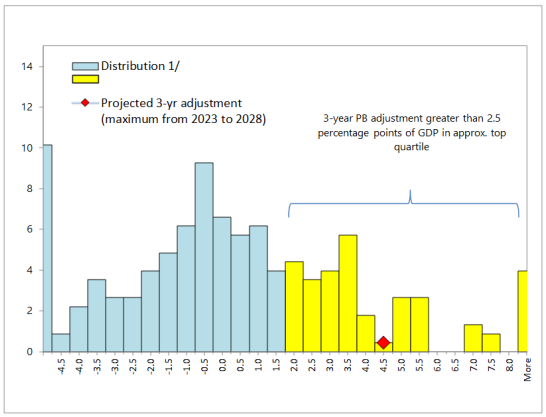
2/ Difference between anticipated and actual contributions on debt ratios.

3/ Distribution across LICs for which LIC DSAs were produced.

4/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

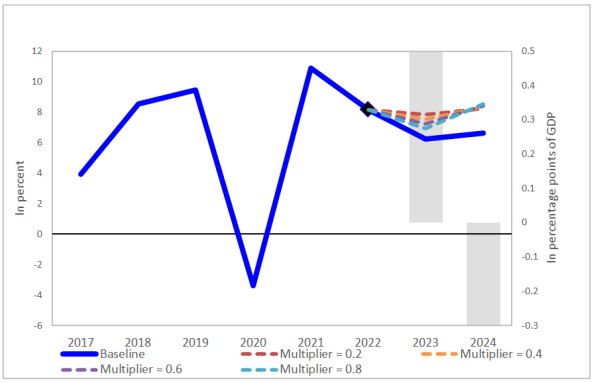
Figure 4. Rwanda: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



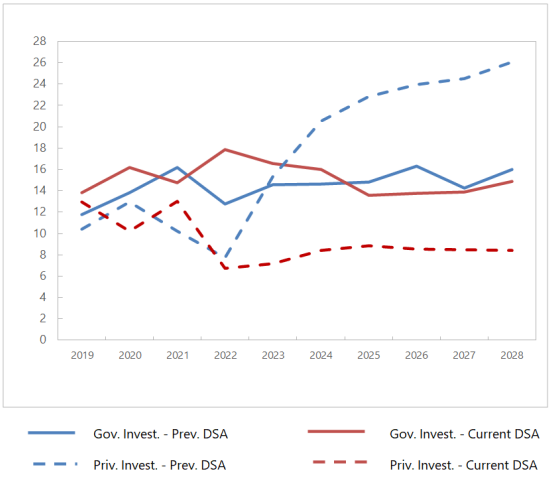
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates 1/
(% of GDP)



1/ Private investment includes the Bugesera airport project.

Contribution to Real GDP growth
(percent, 5-year average)

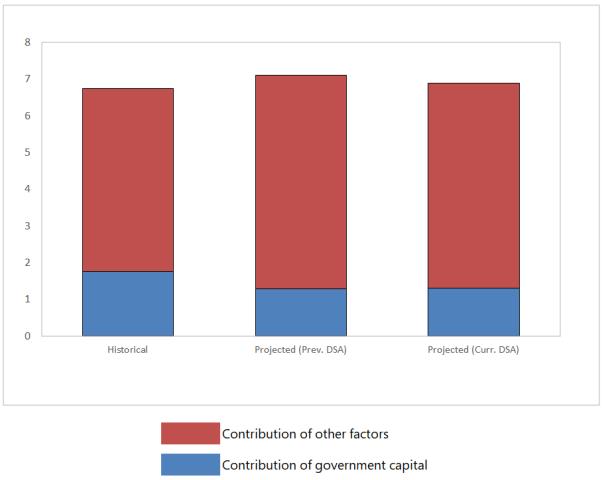


Figure 5. Rwanda: Qualification of the Moderate Category, 2023-33 1/

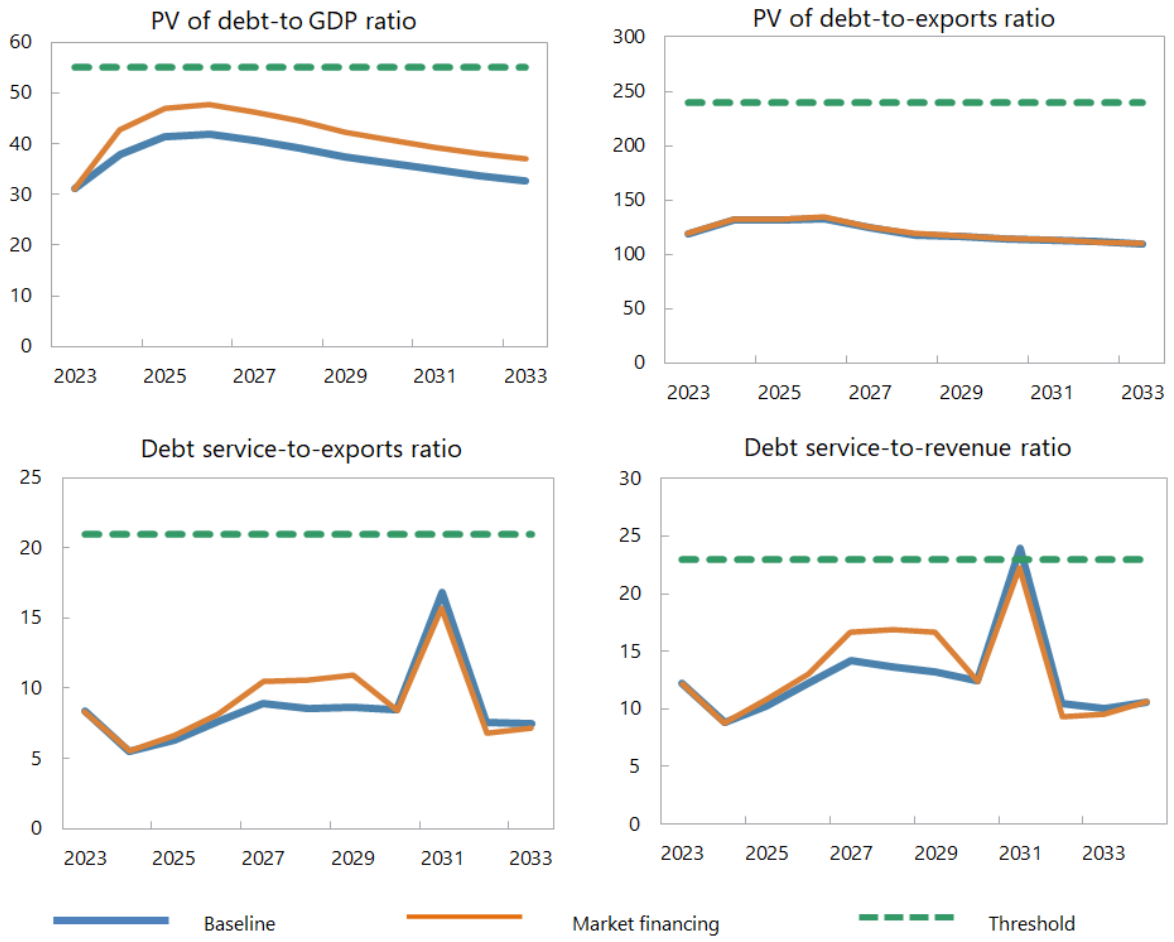
Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

Figure 6. Rwanda: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	13	509
Breach of benchmark	No	No
Potential heightened liquidity needs	Low	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.